Topic 9: Introduction to Kinetics and Equilibria

9A: Kinetics

Students will be assessed on their ability to:

9.1	understand, in terms of the collision theory, the effect of changes in concentration, temperature, pressure and surface area on the rate of a chemical reaction
9.2	understand that reactions take place only when collisions have sufficient energy, known as the activation energy
9.3	be able to calculate the rate of a reaction from: i the time taken for a reaction, using rate = 1/time ii the gradient of suitable graph, by drawing a tangent, either for initial rate, or at a time, t
9.4	understand qualitatively, in terms of the Maxwell-Boltzmann distribution of molecular energies, how changes in temperature affect the rate of a reaction
9.5	understand the role of catalysts in providing alternative reaction routes of lower activation energy
9.6	be able to draw the reaction profiles for uncatalysed and catalysed reactions, including the energy level of the intermediate formed with the catalyst
9.7	understand the use of catalysts in industry to make processes more sustainable by using less energy and/or higher atom economy
9.8	be able to interpret the action of a catalyst in terms of a qualitative understanding of the Maxwell-Boltzmann distribution of molecular energies
	Further suggested practical Experiments to demonstrate the factors that influence the rate of chemical reactions, including the decomposition of hydrogen peroxide, reaction of marble chips with acid, reaction of thiosulfate ions with acid

9B: Equilibria

Students will be assessed on their ability to:

9.9	know that many reactions are readily reversible and that they can reach a state of dynamic equilibrium in which: i the rate of the forward reaction is equal to the rate of the backward reaction ii the concentrations of the reactants and the products remain constant
9.10	be able to predict and justify the qualitative effects of changes of temperature, pressure and concentration on the position of equilibrium in a homogeneous system
9.11	evaluate data to explain the necessity, for many industrial processes, to reach a compromise between the yield and the rate of reaction

	Further suggested practicals: Demonstrate the effect of a change of temperature, pressure and concentration on a system at equilibrium: i chlorine reacting with iodine to form iodine(I) chloride, which then reacts with chlorine to form iodine(III) chloride ii the equilibrium system between nitrogen dioxide (NO_2) and dinitrogen tetroxide (N_2O_4)